

How can companies share internal data safely for organizational research?



John Pumayalli

Symbolic Systems Program, Stanford University • with the Stanford Graduate School of Business under Macro Organizational Behavior
johnpuma@stanford.edu

The open question

- Most of what matters inside a company, its decisions, coordination, priorities, dependencies and changes of direction, has been very hard for researchers to observe systematically.
- Nearly all of it now leaves a digital trace in Slack, meeting transcripts and project tools. Reading that trace by hand is slow, and it is a privacy problem: the raw data is full of names and of the company's own confidential work.
- Research on rich internal data has therefore needed technical infrastructure that only large firms could provide. Startups, both interesting and economically important, were mostly out of reach.
- This introduces the question:** Can a small academic team build a platform that lets a company of any size share its internal data continuously, strips sensitive information before a researcher can see it, and still lets researchers measure what the company and Stanford Institutional Review Board (IRB) agreed to?

Approach: protect first, then analyze

- Nothing sensitive reaches storage.** Connectors pull Slack and meeting transcripts and replace every name and handle with a token ([PERSON_n]) before the first write. The token-to-name mapping is sealed under a key that belongs to that company alone.
- Analysis runs on de-identified data.** The platform summarizes and analyzes the data for organizational analytics: alignment, decision-making, coordination. Researchers work from those measures, not from raw messages.
- Names come back in one place.** Only the company's own dashboard shows real names, restored at the last step, with an audit record for every act of re-identification.
- Everything stays inside one cloud project.** Storage, analysis, the language models and the logs all live in a Google Cloud project the university controls. No text leaves it.

The agent: search and analysis without exposure

- The agent reads the de-identified store from inside the project, under its own database role, with row-level security deciding what it can see.
- It answers with citations to the exact records it read; an answer cannot cite something the turn never observed.
- It has no re-identification capability and will not be given one. Names appear only when a person at the company reads the answer, and that act is logged.

What de-identification looks like

What the company's Slack holds

@Maria Chen asked whether the pricing rollout should slip a week. Alex said sales needs it before the offsite. Decision: ship Friday; Maria owns it.

What reaches storage

[PERSON_1] asked whether the pricing rollout should slip a week. [PERSON_2] said sales needs it before the offsite. Decision: ship Friday; [PERSON_1] owns it.

The organizational content survives intact: a decision, its owner, its deadline, and the dependency behind it. The people do not. These messages are then sent to the LLM pipeline, never viewed by a researcher.

Google Cloud Platform components and how they interact

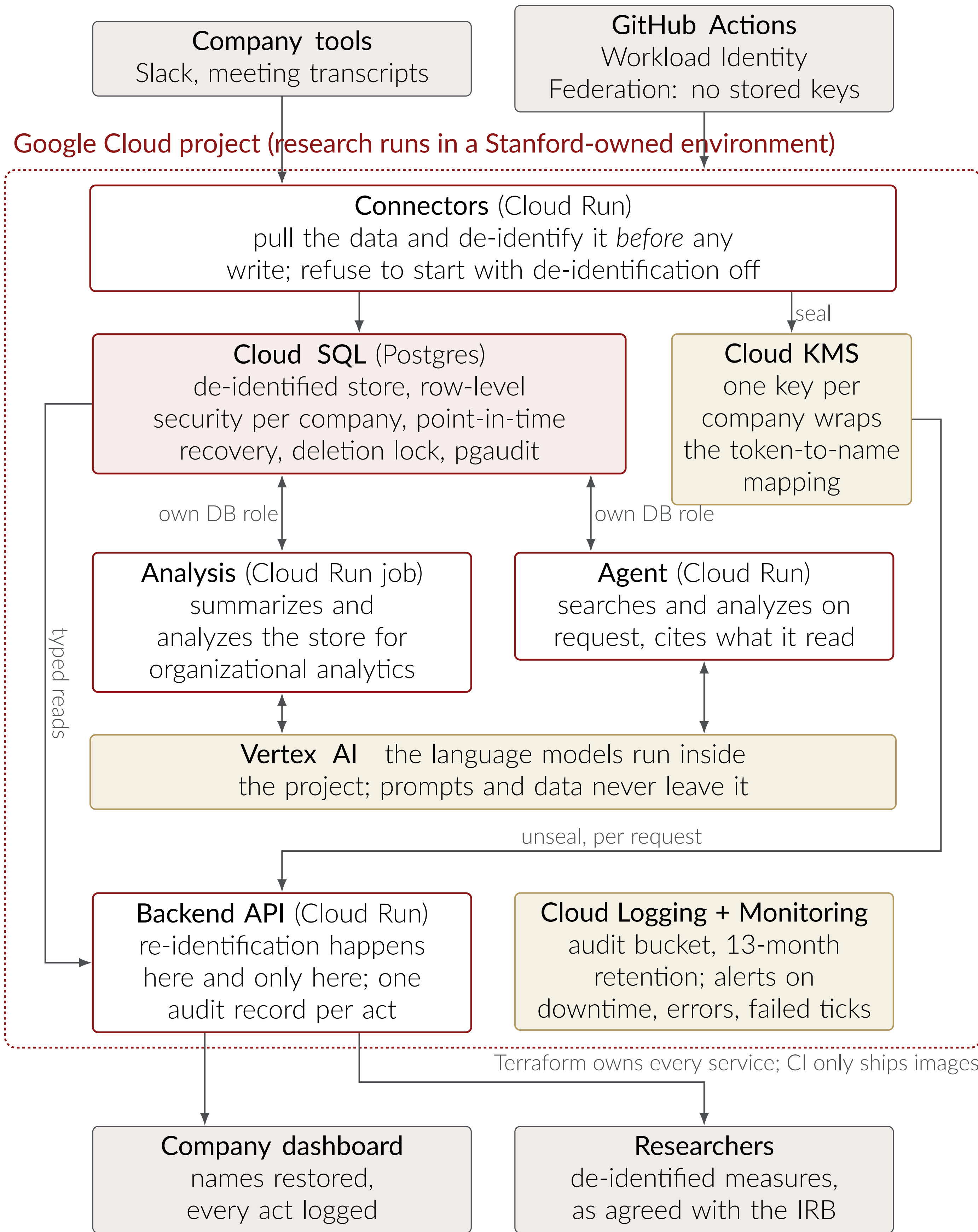


Figure 1. Data crosses the project boundary exactly twice: de-identified on the way in, re-identified (and audited) on the way out to the company's own dashboard.

Open questions

- Which other sources would deepen the picture?** Slack and meetings show how a company talks, outside of which there is still much context left to analyze. Connectors for project trackers, code, documents and calendars would each add a dimension, and each brings its own privacy questions to answer first.
- Can the company become the platform's teacher?** Every summary is a guess until someone who was there confirms it. If the dashboard let people mark what the analysis got right and wrong, each company would be validating its own data, and the system would learn where it is precise and where it drifts, all inside the boundary that keeps researchers out.

What I built

- The research deployment on Stanford's Google Cloud Platform.** The whole platform stood up in a university-owned Google Cloud project, with the language models moved from a third-party API to Vertex AI inside the project.
- SOC 2 controls.** Point-in-time recovery with a rehearsed restore, deletion locks on the database, uptime and error alerting, database audit logging, and data-access logs for every key and secret read.
- Re-identification audit.** Every place the system can turn a token back into a name now goes through one gate that refuses without an identified actor and writes one record per act.
- Nothing holds a key it could leak.** A leaked key is the most common way a cloud system is breached, so this one has none to leak. CI deploys through Workload Identity Federation, which lets Google verify a GitHub workflow directly rather than trusting a stored key. Each service runs as its own account with the narrowest permissions it needs. And a run of the analysis or the agent begins only when the caller presents a Google identity token, so nothing outside the project can trigger one.
- Per-PR preview environments.** Every pull request gets an isolated copy of the full stack, so changes are exercised on realistic data before they reach any shared environment.

What was learned

- Privacy has to be a boundary, not a filter.** De-identifying before storage, rather than redacting on the way out, means there is nothing to leak. The analysis never touches Personally Identifiable Information (PII).
- Citations have to be built into the agent.** Every answer delivered by the queryable agent comes with concrete evidence pointing to messages that are only deanonymized when they reach the user.
- Testing is pivotal, especially when it comes to privacy and data security.** Point-in-time recovery was switched on in an afternoon; proving it took a full drill, restoring the database to a copy and reading the data back. The drill found two settings that would have blocked a real recovery. Had it not been for the test, this block would have gone undetected.
- Keeping the data in one closed environment is the safest option.** Running everything, including the language models, inside a Google Cloud project the university owns changed the privacy question from "whom do we trust" to "what leaves the project, and how".

Why it matters for research

- Companies of any size** can share data through one process, so knowledge about startups no longer depends on a company having an IT team to support the study.
- New measurements.** We now get to study decision velocity, how quickly organizations make and execute decisions, a concept management research has discussed for decades but has had difficulty observing directly in small teams.

The research deployment, by the numbers

0

humans with owner rights on the project

0

long-lived cloud keys in CI or code

1

service that can ever restore a name

13 mo

retention of the audit log